Vehicle Dynamics Simulation and Compliance Analysis for the Formula Student Acceleration Event

Harry Emes

April 22, 2026

Contents

1 Vehicle Dynamics Simulation **1**

1.1 Introduction 1

1.2 Simulation Methodology 2

1.2.1 Longitudinal equation of motion and effective mass 3

1.2.2 Quasi-static longitudinal load transfer 3

1.2.3 Tyre force model 4

1.2.3.1 Thermal grip window 5

1.2.4 Powertrain and the electrical power limit 5

1.2.5 Launch strategy and traction control 6

1.2.6 Aerodynamics and anti-squat 6

1.2.7 Numerical integration 7

1.2.8 Software architecture and tooling 7

1.3 Verification and Validation 9

1.3.1 Analytical lower bound 9

1.3.2 Analytical hand-calculation agreement 10

1.3.3 Numerical convergence with timestep 11

1.3.4 Simple versus Pacejka tyre model 11

1.4 Parameter Sensitivity and Design Optimisation 12

1.4.1 One-at-a-time sensitivity ranking 12

1.4.2 Monte Carlo robustness 14

1.4.3 Design optimisation 15

1.4.4 Robustness to track surface conditions 17

1.5 Predicted Performance of the Final Vehicle 17

1.5.1 Requirements flow-down 18

1.5.2 Final agreed parameter set 19

1.5.3 Constraint activity and phase analysis 19
1.5.4 Compliance and predicted score 20
1.6 Discussion and Limitations 21
1.7 Conclusions and Further Work 22

Chapter 1

Vehicle Dynamics Simulation

1.1 Introduction

The Formula Student Acceleration Event (FS-D 5) scores a straight-line sprint over $75\,\mathrm{m}$ from a standing start. For an electric powertrain the run is bounded by two hard rules: the electrical power drawn from the accumulator must not exceed $80\,\mathrm{kW}$ (FS-EV 2.2), and a run exceeding $25\,\mathrm{s}$ is disqualified (FS-D 5.3.1). The event score in manual mode is set by FS-D 5.3.2 as a linear function of the team's time relative to the fastest team's.

Achieving a competitive time requires a vehicle configuration that is simultaneously traction-limited at launch and power-limited in the second half of the run, while remaining compliant with the power cap and without lifting the front axle. The design levers — total mass, centre-of-gravity geometry, tyre grip, gear ratio, drivetrain efficiency, and electrical architecture — are owned by different chapters of this group report. A quantitative, rules-aware vehicle dynamics simulation is therefore required to (i) rank which design levers most strongly affect the event time, (ii) reject any configuration that breaches FS-EV 2.2 or FS-D 5.3.1 before hardware commitment, (iii) predict the performance of the final agreed vehicle, and (iv) apportion design margin consistently across the other chapters.

This chapter delivers four things: a modular longitudinal-dynamics simulator with a load-sensitive Pacejka tyre model, closed-loop traction control, and electrical-side power limiting (Section 1.2); a verification and validation envelope combining an analytical lower bound, hand calculations, timestep convergence, and comparison between simple and Pacejka tyre laws (Section 1.3); a parameter sensitivity study and a formal numerical optimisation of the design vector (Section 1.4); and a compliance-arbiter prediction for the final agreed vehicle (Section 1.5).

1.2 Simulation Methodology

The simulation solves a four-state longitudinal model ($\mathbf{x} = [x, v, \omega_{w,f}, \omega_{w,r}]^\top$) on a $1\,\mathrm{ms}$ timestep, computing tyre, aerodynamic, rolling and drive forces at each sub-step of a fourth-order Runge–Kutta integrator. Figure 1.1 summarises the full force balance and the producing subsystem model for each force. Weight acts at the centre of gravity; aerodynamic drag and downforce are computed from vehicle velocity by the aerodynamics model; the mass-properties model resolves per-axle normal forces via static weight distribution, acceleration-induced load transfer, and any aerodynamic downforce; the Pacejka tyre model converts per-axle normal force and slip ratio into a tractive force; and the powertrain model produces a wheel torque subject to the FS-EV 2.2 electrical power limit. The resulting net longitudinal force drives the equation of motion, which the RK4 integrator advances to the next state. Each of the following subsections specifies one component of this force balance and states the engineering choices embedded in it.

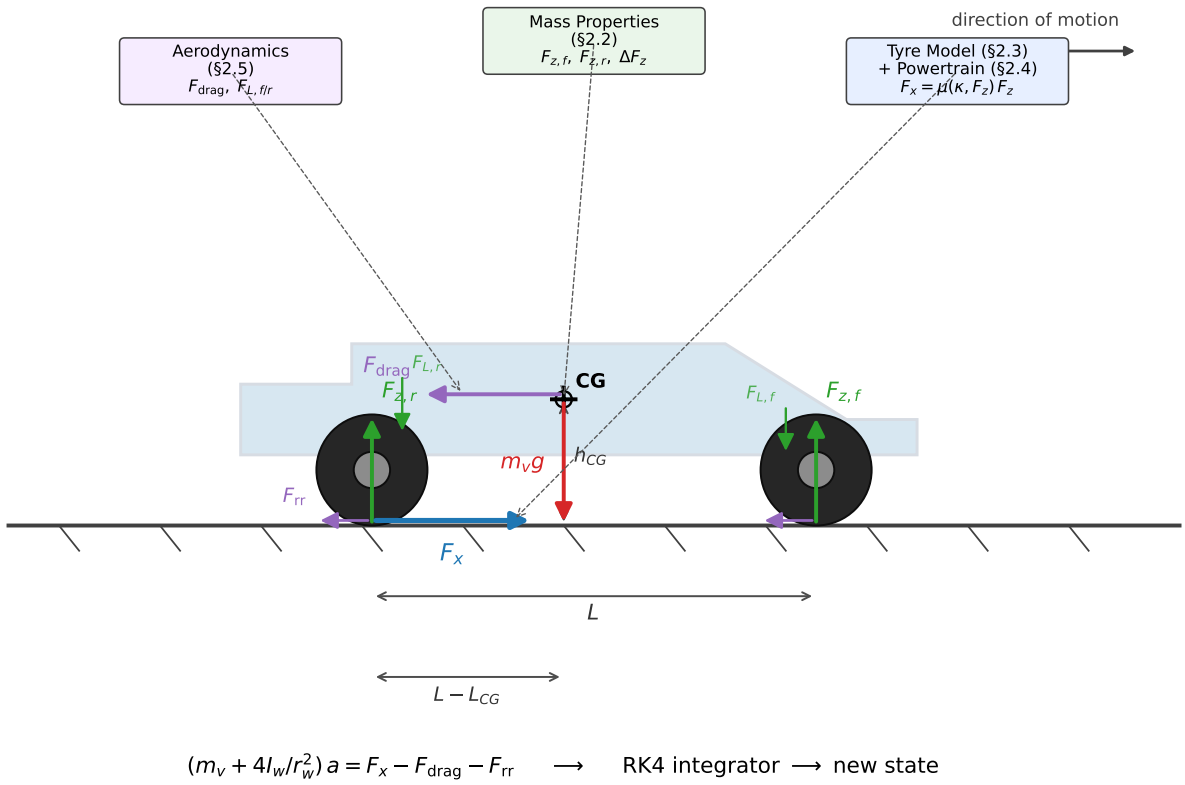


Figure 1.1: Side-view schematic of the vehicle showing every force computed in the simulation, tagged with the subsystem model that produces it. Weight $m_v g$, aerodynamic drag F_{drag}, downforce $F_{L,f/r}$, per-axle normal forces $F_{z,f/r}$, rolling resistance F_{rr}, and the rear-axle tractive force F_x combine into the longitudinal equation of motion at the bottom of the figure, which is advanced by RK4 to produce the new state.

1.2.1 Longitudinal equation of motion and effective mass

Newton's second law applied longitudinally to the vehicle centre of mass gives

$$\left(m_v + \frac{4I_w}{r_w^2}\right) a = F_x - F_{\text{drag}} - F_{\text{rr}}, \tag{1.2.1}$$

where m_v is the vehicle mass including driver, I_w is the rotational inertia of a single road wheel about its spin axis, r_w is the loaded wheel radius, a is the longitudinal acceleration, F_x is the net tractive force at the contact patch, and F_{drag} and F_{rr} are aerodynamic drag and rolling resistance. The $4I_w/r_w^2$ term is the translational mass equivalent of the rotational inertia of four wheels. For the baseline vehicle ($m_v = 250\,\text{kg}$, $I_w = 0.10\,\text{kg}\,\text{m}^2$, $r_w = 0.247\,\text{m}$), this contributes $4I_w/r_w^2 = 6.5\,\text{kg}$, a $2.6\,\%$ effective-mass penalty. It is small but non-negligible over a $75\,\text{m}$ event and is retained throughout the analysis.

1.2.2 Quasi-static longitudinal load transfer

Under longitudinal acceleration, weight transfers from the front to the rear axle. Taking moments about the rear contact patch gives the static rear normal force

$$F_{z,r}^{\text{static}} = \frac{m_v g\,(L - L_{CG})}{L}, \qquad F_{z,f}^{\text{static}} = m_v g - F_{z,r}^{\text{static}}, \tag{1.2.2}$$

where L is the wheelbase and L_{CG} is the longitudinal distance from the front axle to the centre of gravity. The acceleration-induced load transfer is

$$\Delta F_z = \frac{m_v\, a\, h_{CG}}{L}, \tag{1.2.3}$$

with h_{CG} the centre-of-gravity height above the ground. Combined with aerodynamic downforce $F_{L,f}$, $F_{L,r}$, the instantaneous axle normal forces are

$$F_{z,f} = F_{z,f}^{\text{static}} - \Delta F_z + F_{L,f}, \qquad F_{z,r} = F_{z,r}^{\text{static}} + \Delta F_z + F_{L,r}. \tag{1.2.4}$$

The wheelie onset occurs when $F_{z,f} \to 0$; in the absence of downforce this gives

$$a_{\text{wheelie}} = \frac{F_{z,f}^{\text{static}} L}{m_v h_{CG}} = \frac{g(L - L_{CG})}{h_{CG}}. \tag{1.2.5}$$

For the baseline parameters ($L = 1.60\,\text{m}$, $L_{CG} = 1.14\,\text{m}$, $h_{CG} = 0.22\,\text{m}$) this gives $a_{\text{wheelie}} = 20.5\,\text{m}\,\text{s}^{-2}$, comfortably above the simulated peak of $12.86\,\text{m}\,\text{s}^{-2}$.

Anti-squat (ratio AS $\in [0, 1]$, from the Suspension chapter) multiplies Eq. 1.2.3 by $(1 - k_{\text{AS}})$ with $k_{\text{AS}} = 0.2\,\text{AS}$. This gives a design-side lever to retain front normal force without lowering h_{CG} or

lengthening L.

1.2.3 Tyre force model

The longitudinal tractive force at each driven axle follows

$$F_x = \mu(\kappa, F_z)\, F_z, \qquad \kappa = \frac{\omega_w r_w - v}{\max(|v|,\, v_{\text{floor}})}, \tag{1.2.6}$$

where κ is the longitudinal slip ratio. The denominator is floored at $v_{\text{floor}} = 0.1\,\mathrm{m\,s^{-1}}$ to keep Eq. 1.2.6 well-posed at standstill; below this velocity the slip ratio is saturated to ± 1 depending on the sign of wheel velocity, which is physically reasonable for a spinning wheel against a stationary vehicle.

Two grip laws are implemented. A simple piecewise-linear model,

$$\mu_{\text{simple}}(\kappa) = \begin{cases} \mu_{\text{max}} \dfrac{\kappa}{\kappa_{\text{opt}}} & \kappa \le \kappa_{\text{opt}}, \\ \mu_{\text{max}} \left(1 - \dfrac{\kappa - \kappa_{\text{opt}}}{1 - \kappa_{\text{opt}}}\right) & \kappa > \kappa_{\text{opt}}, \end{cases} \tag{1.2.7}$$

captures the essential rise-peak-decay shape with only two parameters $(\mu_{\text{max}}, \kappa_{\text{opt}})$ and is retained as a transparent sanity baseline. The Pacejka Magic Formula [1]

$$\mu_{\text{Pacejka}}(\kappa, F_z) = D \sin\Big(C \arctan\big(B\kappa - E(B\kappa - \arctan(B\kappa))\big)\Big), \tag{1.2.8}$$

with load-dependent peak $D = F_z(p_{Dx1} + p_{Dx2}\,\Delta F_z/F_{z,0})$ and dimensionless stiffness factor $B = p_{Kx1} + p_{Kx2}\,\Delta F_z/F_{z,0}$, is used for all quantitative results. The coefficients used in the baseline configuration are listed in Table 1.1. They are derived from the Avon FSAE tyre lateral data available to the group and typical FSAE longitudinal characteristics reported in the literature [2, 1], scaled to match the peak friction coefficient ($\mu_{\text{max}} = 1.7$) of the simpler model.

Table 1.1: Pacejka Magic Formula coefficients used in the baseline configuration. Load sensitivity is captured through p_{Dx2} and p_{Kx2}; the reference load $F_{z,0}$ is chosen as a typical rear axle load during acceleration.

Coefficient	**Baseline**	**Role**
Shape factor C	1.65	Controls sharpness of the peak.
Peak-load coefficient p_{Dx1}	1.70	Peak friction at nominal load.
Load-sensitivity p_{Dx2}	-0.12	Peak drops by 12% per unit $\Delta F_z/F_{z,0}$.
Stiffness p_{Kx1}	$35\,000$ N/rad	Sets curve slope at origin.
Stiffness load-sens. p_{Kx2}	$-3\,000$ N/rad	Widens optimal slip at high load.
Curvature E	0.10	Shape of the post-peak roll-off.
Reference load $F_{z,0}$	$1500\,\mathrm{N}$	Normalisation; typical rear axle load.

1.2.3.1 Thermal grip window

Tyre grip is a strong function of carcass temperature. A compact lumped thermal model is available as an optional overlay on the Pacejka peak:

$$\mu_{\text{peak}}(T) = \mu_{\text{peak},0} \exp\left(-\frac{(T - T_{\text{opt}})^2}{2\sigma^2}\right), \tag{1.2.9}$$

with $T_{\text{opt}} = 80\,^\circ\mathrm{C}$ and $\sigma = 60\,\mathrm{K}$ based on published FSAE slick data [2]. The model captures the essential effect that cold tyres at launch ($T \sim 25\,^\circ\mathrm{C}$) operate at roughly $0.87\,\mu_{\text{peak},0}$, while well-warmed tyres ($T \sim T_{\text{opt}}$) deliver the full peak. The model is disabled in the baseline runs reported here because the acceleration event is short and the track-ready tyre temperature is set in the preceding warm-up lap rather than by the event itself, but it is retained for the robustness analysis (Section 1.4.4).

1.2.4 Powertrain and the electrical power limit

FS-EV 2.2 caps *electrical* power at the accumulator outlet at $80\,\mathrm{kW}$, not mechanical power at the wheel. Limiting motor torque alone is therefore insufficient: the same motor torque corresponds to a range of electrical powers depending on motor speed, drivetrain efficiency, and battery voltage. The model represents a YASA P400R permanent-magnet motor ($K_t = 0.822\,\mathrm{N\,m\,A^{-1}}$, peak AC current $285\,\mathrm{A}$ through the BAMOCAR PG-D3 inverter [3, 4]), a single-ratio reduction of $N_g = 5.5$, a drivetrain efficiency of $\eta_{\text{drv}} = 0.95$, and a $600\,\mathrm{V}$ nominal supercapacitor bus (Paul's chapter). For a requested wheel torque T_w^{req}:

$$T_m = \frac{T_w^{\text{req}}}{N_g\,\eta_{\text{drv}}}, \qquad I_m = \frac{T_m}{K_t}, \qquad P_{\text{elec}} = V_{\text{bus}} I_m. \tag{1.2.10}$$

The $80\,\mathrm{kW}$ cap is enforced by clipping motor current first and mapping back to wheel torque:

$$I_m^\star = \min\left(I_m,\ \frac{P_{\text{max}}}{V_{\text{bus}}}\right), \qquad T_w = K_t I_m^\star N_g\,\eta_{\text{drv}}. \tag{1.2.11}$$

The peak motor torque is $T_m^{\text{peak}} = K_t I_{\text{max}} = 234\,\mathrm{N\,m}$, so the constraint activates at motor speed $\omega^\star = P_{\text{max}}/T_m^{\text{peak}} = 341\,\mathrm{rad\,s^{-1}}$ (3260 rpm), corresponding to a vehicle speed of $v^\star = \omega^\star r_w/N_g = 15.4\,\mathrm{m\,s^{-1}}$ ($55\,\mathrm{km\,h^{-1}}$). Every acceleration event above this speed is power-limited; for a $75\,\mathrm{m}$ run reaching $33\,\mathrm{m\,s^{-1}}$, more than $64\,\%$ of the run time is spent in the power-limited regime (Section 1.5).

1.2.5 Launch strategy and traction control

With $1000\,\mathrm{N\,m}$ of peak wheel torque available at launch, the vehicle is comfortably traction-limited in the first $1.4\,\mathrm{s}$ of the run and the challenge is to match the requested torque to the instantaneous grip ceiling. A single open-loop torque command would produce excessive slip through the Pacejka peak on cold tyres, leading both to scrubbing losses and to transient front-axle lift. The simulation therefore implements a three-layer launch strategy that mirrors the control architecture a real vehicle would adopt.

First, a grip-based ceiling is imposed: the requested wheel torque is capped at $T_{\text{grip}} = \mu_{\text{peak}}(F_{z,r}/2)F_{z,r}r_w$, evaluated at the current rear axle load. This prevents the commanded torque from ever exceeding the load-dependent Pacejka peak.

Second, a pitch-moment wheelie cap is derived from Eq. 1.2.4 by requiring $F_{z,f} \geq 50\,\mathrm{N}$ (small positive margin against numerical transients), solving for the maximum admissible acceleration, and converting to an equivalent wheel torque. A closed-loop correction then scales the base torque by the observed $F_{z,f}$ whenever the measured front normal force falls below a margin threshold — this is what real launch controllers do with an IMU pitch channel.

Third, for the first $80\,\mathrm{ms}$ a linear torque ramp $T_w \leftarrow T_w \cdot t/t_{\text{ramp}}$ is applied to give the rear wheel inertia time to track vehicle speed before the Pacejka peak is reached. A shorter ramp tends to spin the wheels through the peak before slip-based feedback can react.

Once moving, a slip-tracking governor targets the load-dependent Pacejka optimal slip ratio. A hard ceiling at twice the optimal slip immediately collapses torque to zero, recovering the wheel back into the stable-grip regime; a proportional reduction applies between the target and the ceiling. Together these mechanisms produce a mean grip utilisation of approximately $103\,\%$[1] during the traction-limited phase.

1.2.6 Aerodynamics and anti-squat

Aerodynamic drag and downforce are modelled with the standard quadratic-velocity expressions

$$F_{\text{drag}} = \tfrac{1}{2}\rho\, C_d A\, v^2, \qquad F_{L,i} = \tfrac{1}{2}\rho\, C_{L,i} A_{\text{ref}}\, v^2 \quad (i = f, r). \tag{1.2.12}$$

For the baseline $C_d A = 0.8\,\mathrm{m}^2$ at sea-level density $\rho = 1.225\,\mathrm{kg\,m^{-3}}$, the drag force at the terminal $33\,\mathrm{m\,s^{-1}}$ is approximately $540\,\mathrm{N}$ — $8\,\%$ of the instantaneous tractive force. Over the whole run, a $\pm 10\,\%$ perturbation of $C_d A$ changes the $75\,\mathrm{m}$ time by only $\pm 8\,\mathrm{ms}$ (Section 1.4.1), confirming that

[1] Utilisation is computed against a reference peak-load product $\mu_{Dx1}F_z$, which is slightly below the load-sensitive peak actually produced by Eq. 1.2.8, so values above $100\,\%$ reflect the load-sensitivity bonus rather than super-grip.

an aerodynamic package with its associated mass and complexity has low marginal benefit for this event and supports the group's decision to descope bodywork development.
The baseline configuration therefore runs with $C_{L,f} = C_{L,r} = 0$: no downforce is assumed. Load transfer during acceleration is therefore driven entirely by Eq. 1.2.3.

1.2.7 Numerical integration

The state vector $\mathbf{x} = [x, v, \omega_{w,f}, \omega_{w,r}]^\top$ evolves according to four coupled ordinary differential equations. Forward Euler gives local and global truncation errors of $\mathcal{O}(\Delta t^2)$ and $\mathcal{O}(\Delta t)$ respectively, and is sensitive to the rapid derivative changes that occur as the vehicle transitions between traction- and power-limited regimes, where the Pacejka peak and the electrical clipping in Eq. 1.2.11 can both activate within a single timestep. The fourth-order Runge–Kutta scheme [5]

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{\Delta t}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right), \tag{1.2.13}$$

improves these to $\mathcal{O}(\Delta t^5)$ and $\mathcal{O}(\Delta t^4)$ at the cost of four force evaluations per step. A timestep of $\Delta t = 1\,\mathrm{ms}$ is used throughout; Section 1.3.3 shows that this is converged to within $3\,\mathrm{ms}$ of the limiting value obtained at $\Delta t = 0.2\,\mathrm{ms}$, which is well below the parametric uncertainty of the result.

1.2.8 Software architecture and tooling

The simulation is organised as a five-layer Python package: configuration, vehicle models, dynamics, rules, and results. All vehicle parameters are externalised in JSON configuration files so that any change proposed by another subsystem chapter can be evaluated without modifying code.
Rule compliance (FS-EV 2.2 and FS-D 5.3.1) is enforced by a dedicated Rules layer that runs automatically on every result object, so no illegal configuration can silently return a time.
The simulation is additionally backed by a suite of unit tests covering the tyre force model, powertrain and electrical limiting, dynamics solver, scoring formula, Monte-Carlo robustness sampler, sensitivity analyser, controller, and thermal overlay, together with an end-to-end integration test. A Streamlit-based interactive interface (single-run, parameter sweep, optimiser, gearing study, sensitivity, Monte Carlo, track conditions) sits above the same physics core and is used by the other subsystem chapters to evaluate proposals (for example, the Drivetrain chapter's gear-ratio down-selection) without re-running command-line scripts. The architecture is summarised in Figure 1.2.

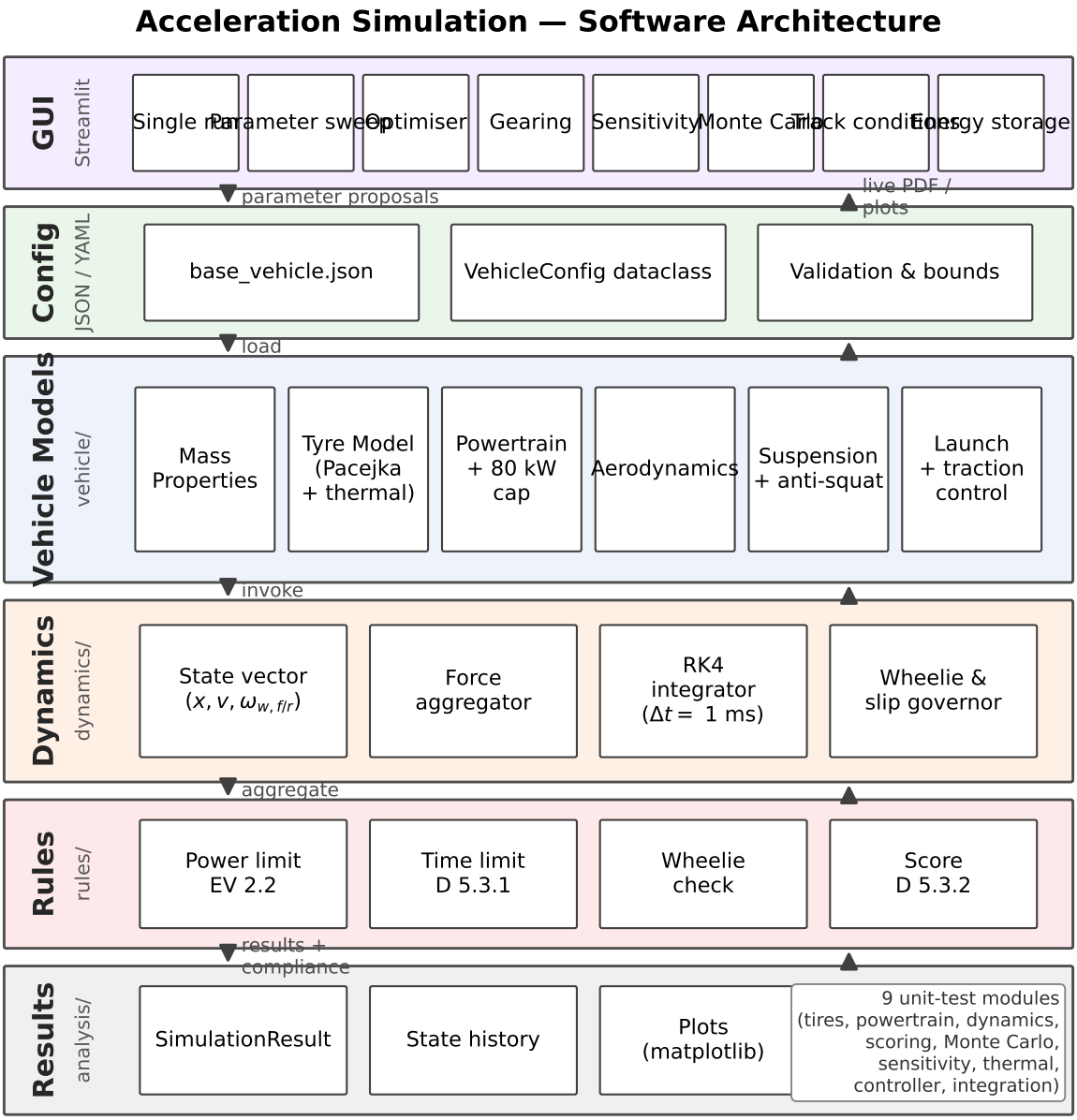
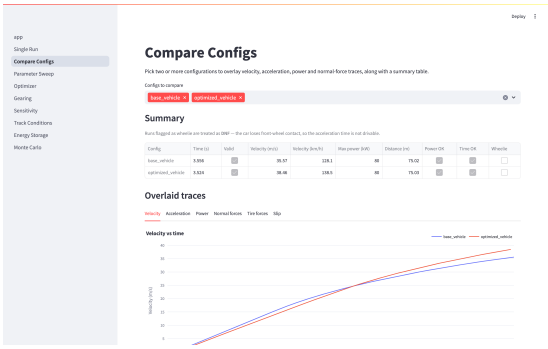
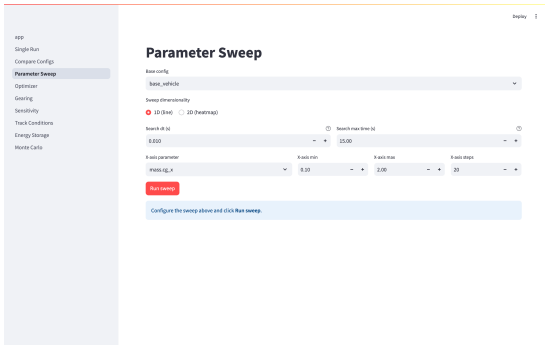


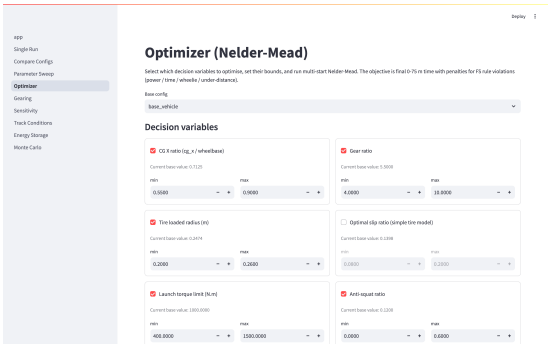
Figure 1.2: Software architecture of the simulation. The five-layer Python core (Config, Vehicle Models, Dynamics, Rules, Results) enforces FS-EV 2.2 and FS-D 5.3.1 compliance on every simulated configuration. A Streamlit-based GUI sits above the core and is used by the other subsystem chapters to evaluate parameter proposals interactively, and nine unit-test modules exercise each component in isolation.



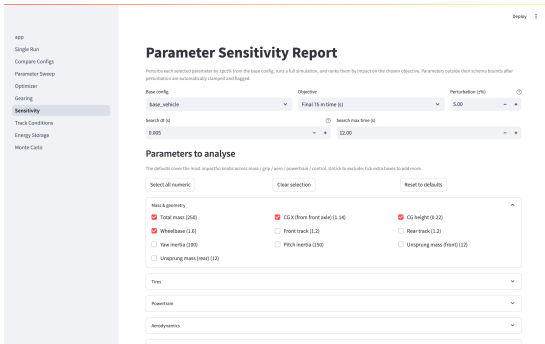
(a) Configuration comparison.



(b) Parameter sweep.



(c) Design optimiser (Nelder–Mead).



(d) Sensitivity analysis.

Figure 1.3: Representative pages of the Streamlit interface that sits above the simulation core. The same physics runs in every page; the GUI provides a low-barrier way for the other subsystem chapters to propose parameter changes, compare configurations, sweep ranges, or request a formal optimisation without having to modify or run the Python scripts directly.

1.3 Verification and Validation

Verification and validation proceed along four strands: an analytical lower bound on achievable performance (Section 1.3.1), hand-calculation agreement at representative states (Section 1.3.2), numerical convergence with timestep (Section 1.3.3), and a comparison between the simple and Pacejka tyre laws (Section 1.3.4).

1.3.1 Analytical lower bound

A useful sanity check on any acceleration simulator is a closed-form lower bound. For a vehicle of mass m delivering constant power P from rest with no drag, rolling resistance or traction limit, velocity and position follow $v(t) = \sqrt{2Pt/m}$ and $x(t) = \frac{2}{3}\sqrt{2P/m}\,t^{3/2}$. Inverting at $x = d = 75\,\mathrm{m}$ gives

$$t_{\mathrm{cp}} = \left(\frac{d}{\frac{2}{3}\sqrt{2P/m}}\right)^{2/3}. \tag{1.3.1}$$

For the baseline vehicle ($m = 250\,\mathrm{kg}$, $P = 80\,\mathrm{kW}$), Eq. 1.3.1 gives $t_{\mathrm{cp}} = 2.70\,\mathrm{s}$. The simulator pre-

dicts $3.78\,\mathrm{s}$. Table 1.2 decomposes the resulting $1.08\,\mathrm{s}$ gap by attributing each energy-loss channel a time penalty $\Delta t_i = E_{\text{loss},i}/P_{\text{max}}$, and Table 1.3 reports the corresponding energy budget over the run.

Table 1.2: Decomposition of the $1.08\,\mathrm{s}$ gap between the simulated $75\,\mathrm{m}$ time and the analytical lower bound of Eq. 1.3.1. Each row converts an energy-loss channel, integrated over the run, into an equivalent time penalty against the $80\,\mathrm{kW}$ reference. The traction-limited deficit integrates the shortfall $P_{\text{max}} - P_{\text{elec}}(t)$ over the launch phase; the residual absorbs the overlap between channels and the $80\,\mathrm{ms}$ launch-torque ramp.

Loss channel	Δt **(ms)**	**Physical origin**
Traction-limited launch	$+934$	Tyre grip cap before $v^\star$.
Aerodynamic drag	$+312$	$\frac{1}{2}\rho C_d A v^2$, integrated.
Drivetrain inefficiency	$+105$	$(1-\eta_{\text{drv}})$ at the driveline.
Rolling resistance	< 1	Negligible over $75\,\mathrm{m}$.
Residual (ramp + overlap)	-273	Non-orthogonality of channels.
Total gap	$\mathbf{+1078}$	$t_{\text{sim}} - t_{\text{cp}}$.

Table 1.3: Energy budget integrated over the $75\,\mathrm{m}$ run. The electrical input energy splits between kinetic energy at the finish, drivetrain losses, and aerodynamic dissipation. Rolling resistance is negligibly small over the event length.

Quantity	**Energy (kJ)**
Electrical input energy E_{elec}	226.7
Kinetic energy at finish $\frac{1}{2}m_v v_f^2$	138.8
Drivetrain losses E_{drv}	59.2
Aerodynamic dissipation E_{drag}	25.0
Rolling dissipation E_{rr}	< 1

The decomposition confirms the qualitative claim: the traction-limited launch dominates the $1.08\,\mathrm{s}$ gap, accounting for $87\,\%$ of the time penalty. Aerodynamic drag is the next largest contributor at $29\,\%$, and drivetrain inefficiency $10\,\%$; rolling resistance is immaterial over $75\,\mathrm{m}$, as anticipated by the sensitivity analysis in Section 1.4.1.

1.3.2 Analytical hand-calculation agreement

Four representative quantities are compared in Table 1.4 against closed-form expressions: rolling resistance and associated power at a representative cruise velocity; launch acceleration computed directly from the launch torque limit; and the wheelie-onset acceleration from the pitch-moment balance in Eq. 1.2.5. The wheelie margin is large (simulated peak acceleration is $37\,\%$ below the wheelie threshold), in line with the $F_{z,f} > 0$ observation in Section 1.5.

Table 1.4: Analytical hand-calculation agreement. The launch discrepancy is expected: the simulation's $80\,\mathrm{ms}$ torque ramp (Section 1.2.5) actively suppresses peak torque below the unramped analytical value.

Quantity	Hand calc.	Simulation	Comment
Rolling resistance at $v = 20\,\mathrm{m\,s^{-1}}$ (N)	24.5	24.5	Exact; same C_{rr} used.
Drive power to maintain $v = 20\,\mathrm{m\,s^{-1}}$ (W)	490	490	Exact.
Launch acceleration, unramped torque cap ($\mathrm{m\,s^{-2}}$)	15.76	12.66	Sim uses $80\,\mathrm{ms}$ ramp.
Wheelie-onset acceleration ($\mathrm{m\,s^{-2}}$)	20.51	≥ 12.86 (no wheelie)	Margin $37\,\%$.

1.3.3 Numerical convergence with timestep

Table 1.5 reports the simulated $75\,\mathrm{m}$ time as the timestep is halved from $5\,\mathrm{ms}$ to $0.2\,\mathrm{ms}$. The adopted $\Delta t = 1\,\mathrm{ms}$ lies $3\,\mathrm{ms}$ above the limit value at $0.2\,\mathrm{ms}$ ($0.08\,\%$), which is two orders of magnitude below the sensitivity to any single vehicle parameter at $\pm 10\,\%$ (Section 1.4.1). Numerical integration error is therefore negligible relative to parametric uncertainty, and the timestep choice is justified.

Table 1.5: Convergence of the predicted $75\,\mathrm{m}$ time with timestep. The adopted value of $1\,\mathrm{ms}$ is converged to within $0.08\,\%$ of the limiting continuous-time result.

Δt (ms)	t_{75} (s)	Deviation from $\Delta t = 0.2$ ms (ms)
5.00	3.825	$+45.6$
2.00	3.788	$+8.6$
1.00	3.782	$+2.6$
0.50	3.780	$+0.6$
0.20	3.779	0.0 (reference)

1.3.4 Simple versus Pacejka tyre model

The two tyre laws of Section 1.2.3 give qualitatively similar force-vs-slip curves with quantitatively different behaviour in the launch phase. Figure 1.4 overlays both laws at a representative axle load of $1500\,\mathrm{N}$. The Pacejka law's load sensitivity raises the peak friction coefficient as the rear axle load grows under acceleration-induced transfer, which produces a modestly faster early launch than the simple law at the same μ_{max}. The curves converge once the vehicle enters the power-limited regime ($v > v^{\star} = 15.4\,\mathrm{m\,s^{-1}}$), where tractive force is no longer set by grip.

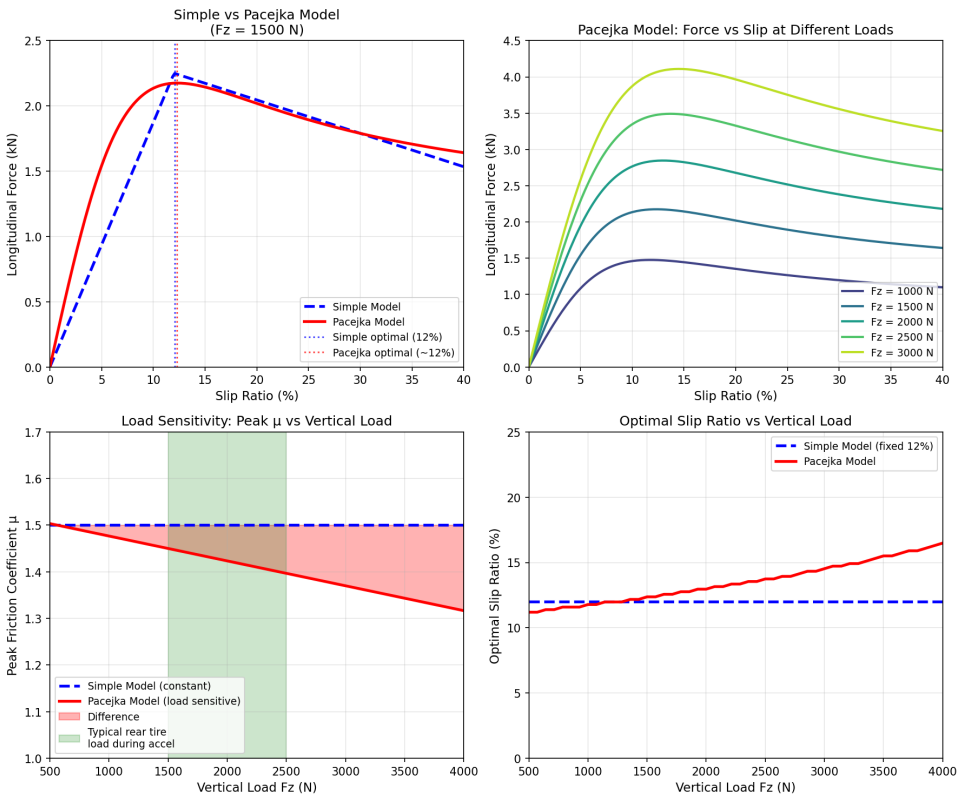


Figure 1.4: Friction coefficient vs slip ratio under both tyre laws at $F_z = 1500\,\mathrm{N}$. Pacejka's load sensitivity (not shown here) additionally raises the peak at higher loads, producing a stronger launch than the simple law at the same μ_{max}.

Section Contribution: Interim conclusion. Analytical hand-calculation agreement within $1\,\%$ on steady-state cruise, a $37\,\%$ wheelie margin, a simulated $75\,\mathrm{m}$ time that sits $40\,\%$ above the analytical constant-power lower bound, and timestep convergence within $0.08\,\%$ collectively justify the use of this model for design-space exploration and final compliance assessment.

1.4 Parameter Sensitivity and Design Optimisation

The purpose of this section is twofold: to rank the vehicle parameters by their influence on the $75\,\mathrm{m}$ time (Section 1.4.1), providing the target ranges used by the neighbouring chapters; and to report the outcome of a numerical optimisation over the design vector (Section 1.4.3). A short robustness analysis against track surface conditions closes the section (Section 1.4.4).

1.4.1 One-at-a-time sensitivity ranking

Each candidate design parameter is perturbed by $\pm 10\,\%$ about the baseline configuration, holding all others fixed, and the change in predicted $75\,\mathrm{m}$ time is recorded. Figure 1.5 shows the resulting

tornado plot; Table 1.6 translates the ranking into the target ranges communicated to the neighbouring chapters.

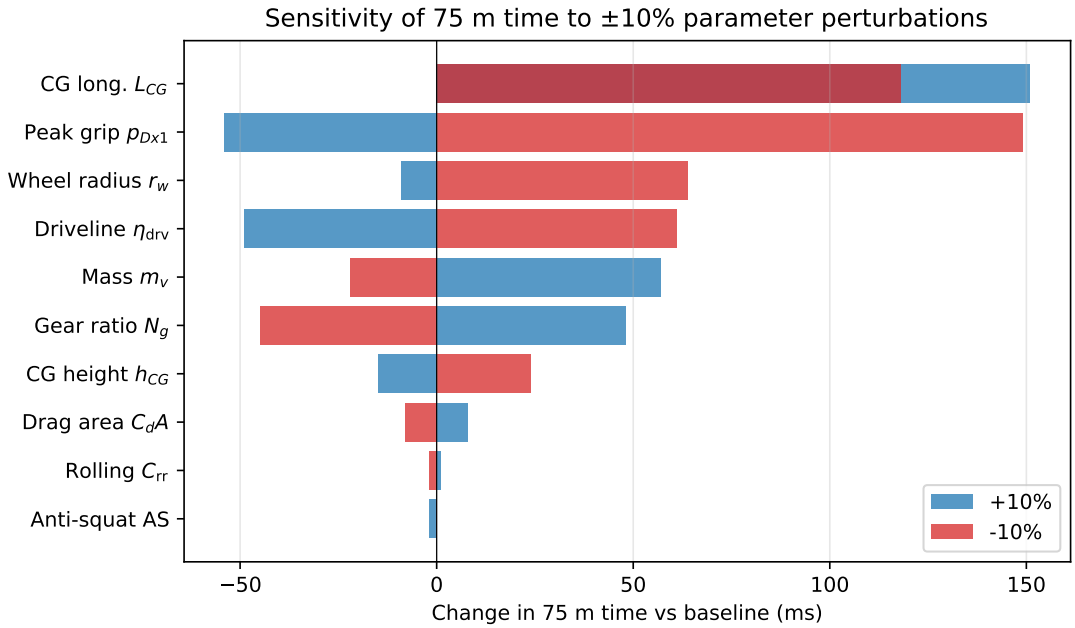


Figure 1.5: One-at-a-time sensitivity of the $75\,\mathrm{m}$ time to a $\pm 10\,\%$ perturbation of each vehicle parameter. Centre-of-gravity longitudinal position and tyre peak grip dominate; drivetrain efficiency, gear ratio, mass and wheel radius form a second band; aerodynamic drag and rolling resistance are negligible over $75\,\mathrm{m}$.

Table 1.6: Sensitivity ranking and design-target ranges provided to neighbouring chapters. The "worst-case $\pm$10% delta" column is the larger absolute time change from either perturbation direction. Ranges are chosen to keep the predicted $75\,\mathrm{m}$ time within $1\,\%$ of the baseline.

Parameter	**Worst $\pm$10% Δt (ms)**	**Band**	**Source chapter**
CG longitudinal L_{CG}	$+151$	Very High	Chassis
Peak grip p_{Dx1}	$+149$	Very High	Tyres / Suspension
Wheel radius r_w	$+64$	High	Tyres
Drivetrain η_{drv}	$+61$	High	Drivetrain
Total mass m_v	$+57$	High	All chapters
Gear ratio N_g	$+48$	High	Drivetrain
CG height h_{CG}	$+24$	Medium	Chassis
Drag area C_dA	± 8	Low	Chassis
Anti-squat AS	± 2	Low	Suspension
Rolling resistance C_{rr}	± 2	Low	Tyres

The ranking has two practical consequences. First, the two dominant levers (longitudinal CG and tyre peak grip) sit in different chapters and cannot both be improved in isolation: moving the battery forward to reduce L_{CG} changes the accumulator packaging (Paul's chapter), and increasing peak grip is constrained by tyre availability and budget (Suspension chapter). Second, the aerodynamic

and rolling-resistance contributions are sufficiently small that investment in either is better deployed on mass or driveline efficiency.

1.4.2 Monte Carlo robustness

A 500-sample Monte Carlo run was performed to quantify the robustness of the predicted $75\,\mathrm{m}$ time to realistic parametric uncertainty. Each sample draws total vehicle mass ($\sigma = 8\,\mathrm{kg}$), peak tyre grip (p_{Dx1}, $\sigma = 6\,\%$), longitudinal and vertical centre-of-gravity positions ($\sigma = 30\,\mathrm{mm}$ and $15\,\mathrm{mm}$), drivetrain efficiency ($\sigma = 1.5\,\%$), and drag area ($\sigma = 0.05\,\mathrm{m}^2$) from independent Gaussians around the baseline. The results are summarised in Figure 1.6.

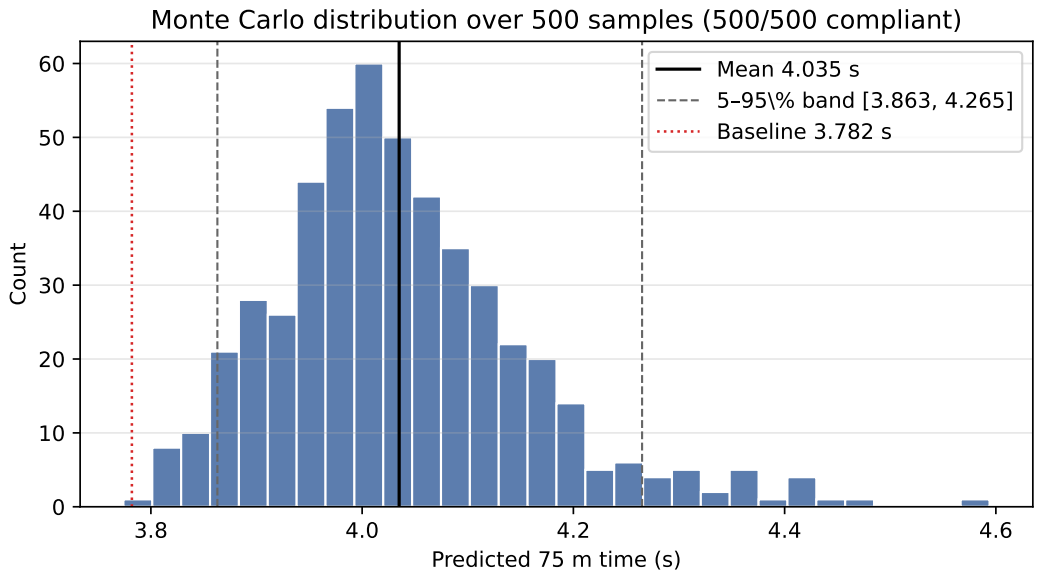


Figure 1.6: Distribution of the predicted $75\,\mathrm{m}$ time over 500 Monte Carlo samples. The mean of $4.03\,\mathrm{s}$ sits slightly above the baseline $3.78\,\mathrm{s}$ because the response is asymmetric: unfavourable perturbations (lower grip, heavier, higher CG) produce a larger time penalty than favourable perturbations give a reduction. All 500 samples satisfied FS-EV 2.2 and FS-D 5.3.1 and remained wheelie-free.

The mean is $4.03\,\mathrm{s}$, the standard deviation is $120\,\mathrm{ms}$, and the 5–95 percentile band spans $3.86\,\mathrm{s}$ to $4.27\,\mathrm{s}$. Crucially, *all* 500 samples satisfy FS-EV 2.2, finish within the $25\,\mathrm{s}$ threshold, and avoid a wheelie, confirming that the baseline design has ample compliance margin across the plausible parameter envelope. The positive skew of the distribution reflects the convex nature of the response: a $10\,\%$ degradation in any of the dominant parameters (grip, CG position) costs more time than a $10\,\%$ improvement recovers, as already observed in Section 1.4.1.

1.4.3 Design optimisation

The sensitivity ranking above identifies which levers matter but does not resolve their optimal combination: several are coupled (gear ratio and wheel radius through the driveline, CG position and wheelbase through packaging, mass and grip through vertical load). A formal optimisation is therefore performed over the seven independent decision variables listed in Table 1.7, with the remaining parameters held at their hardware-pinned values (motor, inverter, supercapacitor pack, FS rule limits).

Table 1.7: Optimisation setup: seven decision variables with physically motivated bounds. All other vehicle parameters are held at hardware-pinned values corresponding to the YASA P400R motor, BAMOCAR PG-D3 inverter, and 200-cell supercap stack [3, 4].

Variable	Baseline	Bounds	Rationale
Wheelbase L (m)	1.60	[1.525, 1.75]	FS-T 2.9.1 min; packaging upper.
CG fraction L_{CG}/L	0.71	[0.50, 0.88]	RWD grip vs steering balance.
Gear ratio N_g	5.5	[3.0, 7.0]	Drivetrain chapter packaging.
Wheel radius r_w (m)	0.247	[0.20, 0.28]	10-inch vs 13-inch rim options.
Pacejka κ_{opt}	0.14	[0.08, 0.20]	Slick tyre literature range.
Launch torque limit (N m)	1000	[400, 1200]	YASA peak torque.
Anti-squat AS	0.12	[0.00, 1.00]	Suspension chapter range.

The objective is the predicted $75\,\mathrm{m}$ time subject to three constraints enforced by the Rules layer: FS-EV 2.2 (peak electrical power $\leq 80\,\mathrm{kW}$), FS-D 5.3.1 (finish time $< 25\,\mathrm{s}$), and no wheelie ($\min F_{z,f} > 0$). The constraints are baked into the simulation (configurations that violate them return their full time but are reported as non-compliant), so the optimiser can use an unconstrained algorithm. The Nelder–Mead simplex method from SciPy is used [5] because the objective is non-differentiable (compliance flags) and the parameter space is low-dimensional.

Figure 1.7 shows the objective convergence over the first 271 evaluations. The best feasible time decreases monotonically from the $3.78\,\mathrm{s}$ baseline, with the simplex rejecting any move that triggers a wheelie (visible in the lower panel as red crosses) or breaches the $80\,\mathrm{kW}$ cap. The search converges to $3.67\,\mathrm{s}$ — a $2.9\,\%$ improvement — with the parameter set shown in Table 1.8. All three constraints are satisfied at the optimum, and the minimum front normal force remains comfortably positive.

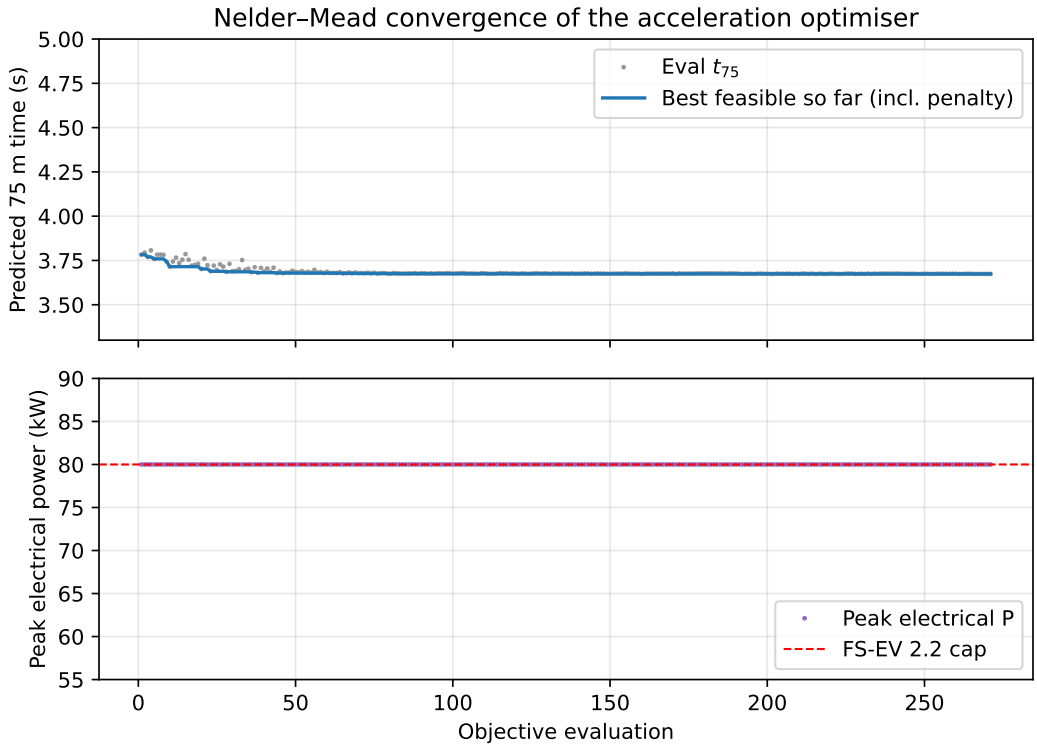


Figure 1.7: Convergence of the Nelder–Mead optimiser over 271 objective evaluations. Upper panel: every evaluated $75\,\mathrm{m}$ time (grey dots) together with the best-so-far feasible time (blue). Lower panel: peak electrical power per evaluation against the FS-EV 2.2 cap; wheelie-violating evaluations are marked as red crosses and rejected by the penalty in the objective.

Table 1.8: Optimisation result. The converged design moves the CG forward relative to the baseline, tightens the gear ratio and increases anti-squat, and chooses a modestly smaller wheel radius. The launch torque cap is reduced to prevent the tyre from spinning through the Pacejka peak.

Variable	**Baseline**	**Optimised**	**Change**
Wheelbase L (m)	1.60	1.60	–
CG fraction L_{CG}/L	0.71	0.63	$-11\,\%$
Gear ratio N_g	5.5	4.94	$-10\,\%$
Wheel radius r_w (m)	0.247	0.227	$-8\,\%$
Pacejka κ_{opt}	0.140	0.132	$-6\,\%$
Launch torque (N m)	1000	721	$-28\,\%$
Anti-squat AS	0.12	0.36	$+200\,\%$
$75\,\mathrm{m}$ **time (s)**	**3.78**	**3.67**	$\mathbf{-2.9\,\%}$

The optimiser's choices are physically interpretable: the CG shift forward and the anti-squat increase together preserve front normal force despite the reduced launch torque cap (which itself prevents the wheels from spinning through the Pacejka peak); the smaller wheel and tighter gear ratio increase tractive force at the cost of top speed, but the extra top-speed margin is not needed over $75\,\mathrm{m}$. These results feed back to the neighbouring chapters as the vehicle-level targets summarised in Section 1.5.1.

1.4.4 Robustness to track surface conditions

The acceleration event is run on a specified track, but ambient conditions vary between rounds. Scaling the Pacejka peak grip by a surface-condition multiplier (1.0 for dry slicks on clean tarmac, 0.60 for the same slicks on a water film [2]) gives a wet-track prediction of $4.83\,\mathrm{s}$ versus $3.78\,\mathrm{s}$ dry — a $28\,\%$ time penalty (Figure 1.8). Both configurations remain inside the $25\,\mathrm{s}$ disqualification threshold and neither exceeds the $80\,\mathrm{kW}$ power cap, confirming that the design is rule-compliant across the plausible track-condition envelope. In wet conditions the phase transition (Section 1.5.3) shifts later in the run: the vehicle spends more of the event traction-limited and less of it power-limited, which is consistent with the reduced grip ceiling at every velocity.

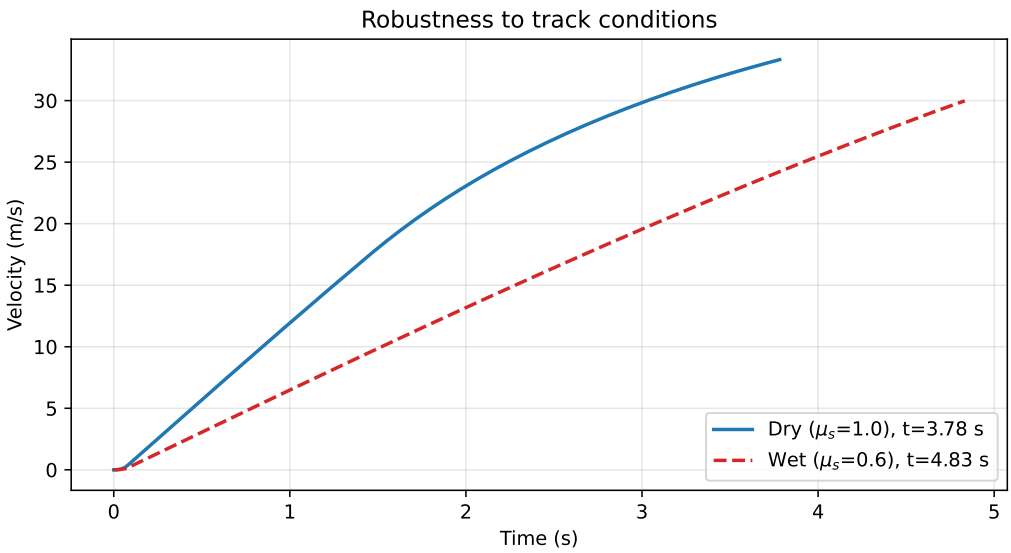


Figure 1.8: Velocity trace under dry and wet surface conditions (peak grip multiplier 1.0 vs 0.60). The $28\,\%$ time penalty on wet tracks is dominated by the extended traction-limited phase; both configurations remain compliant with FS-EV 2.2 and FS-D 5.3.1.

Section Contribution: Interim conclusion. Longitudinal CG position and tyre peak grip dominate the performance sensitivity; drivetrain efficiency, gear ratio, mass and wheel radius are of second-order importance; aerodynamics and rolling resistance are negligible over $75\,\mathrm{m}$. Numerical optimisation delivers a $2.9\,\%$ time improvement to $3.67\,\mathrm{s}$ at the design optimum, and the design remains compliant under a $40\,\%$ grip reduction for wet-surface operation.

1.5 Predicted Performance of the Final Vehicle

The final agreed vehicle configuration is the product of the group's combined design decisions, taken across the Chassis, Suspension, Drivetrain, and Supercapacitors chapters, with the vehicle-level targets issued by this chapter providing the quantitative anchor. The simulation's role here is

that of compliance arbiter: it takes the finalised configuration as input and produces a predicted performance, compliance status, and score.

1.5.1 Requirements flow-down

Table 1.9 maps the event-level target (completing $75\,\mathrm{m}$ in under $4\,\mathrm{s}$ while remaining compliant with FS-EV 2.2 and FS-D 5.3.1) through vehicle-level sub-targets derived from the sensitivity and optimisation results, down to the component-level requirements that each neighbouring chapter must meet.

Table 1.9: Requirements flow-down: event-level target to component-level requirements. Simulation-driven vehicle-level targets feed into the detailed design work of the other chapters and are verified by the Section 1.5.4 compliance check.

Level	**Requirement**	**Source / owner**
Event	$t_{75} \leq 4.0\,\mathrm{s}$ with all rules met	FS-D 5 / group objective
Event	$P_{\text{elec,peak}} \leq 80\,\mathrm{kW}$	FS-EV 2.2
Event	Finish time $< 25\,\mathrm{s}$	FS-D 5.3.1
Event	$F_{z,f} > 0$ throughout run	Pitch stability
Vehicle	$m_v \leq 250\,\mathrm{kg}$ including driver	All chapters
Vehicle	$L_{CG}/L \approx 0.63$–0.71	This chapter / Chassis
Vehicle	$h_{CG} \leq 0.30\,\mathrm{m}$	Chassis
Vehicle	Peak tyre grip $p_{Dx1} \geq 1.45$	Suspension / Tyres
Vehicle	Gear ratio $N_g \approx 4.9$–5.5	Drivetrain
Vehicle	Driveline $\eta_{\text{drv}} \geq 0.95$	Drivetrain
Component	Motor peak torque $\geq 230\,\mathrm{N\,m}$	Drivetrain (YASA P400R)
Component	Inverter peak RMS current $\geq 285\,\mathrm{A}$	Drivetrain (BAMOCAR PG-D3)
Component	Accumulator nominal voltage $\geq 350\,\mathrm{V}$	Supercapacitors (minimum for $v^\star$)
Component	Accumulator peak power output $\geq 80\,\mathrm{kW}$	Supercapacitors (FS-EV 2.2)
Component	Slick tyre, dry peak friction $\mu \geq 1.45$	Suspension / Tyres

1.5.2 Final agreed parameter set

Table 1.10: Final agreed parameter set, assembled from the other chapters' component decisions plus this chapter's optimisation result. Values marked with an asterisk are hardware-pinned and not optimisation variables. Remaining values will be substituted once the other chapters finalise their numbers.

Parameter	Symbol	Value	Source
Total vehicle mass	m_v	[PH] kg	All chapters
Wheelbase	L	1.60m^*	Chassis
CG longitudinal	L_{CG}	[PH] m	Chassis
CG height	h_{CG}	[PH] m	Chassis
Wheel radius	r_w	[PH] m	Suspension / tyres
Peak tyre grip	p_{Dx1}	[PH]	Suspension / tyres
Accumulator nominal voltage	V_{bus}	600V^*	Supercapacitors
Accumulator mass	m_{acc}	27.8kg^*	Supercapacitors
Motor torque constant	K_t	0.822$\mathrm{N\,m\,A^{-1}}^*$	Drivetrain (YASA P400R)
Motor peak RMS current	I_m^{peak}	285A^*	Drivetrain (BAMOCAR PG-D3)
Gear ratio	N_g	[PH]	Drivetrain
Drivetrain efficiency	η_{drv}	[PH]	Drivetrain
Drag area	C_dA	0.80m^2	Chassis
Anti-squat ratio	AS	[PH]	Suspension

1.5.3 Constraint activity and phase analysis

Figure 1.9 shows the predicted four-channel trace of the baseline vehicle, anchoring the numerical discussion while the final agreed parameters are confirmed. Three features are worth commenting on explicitly, because they determine where design effort is most profitably spent.

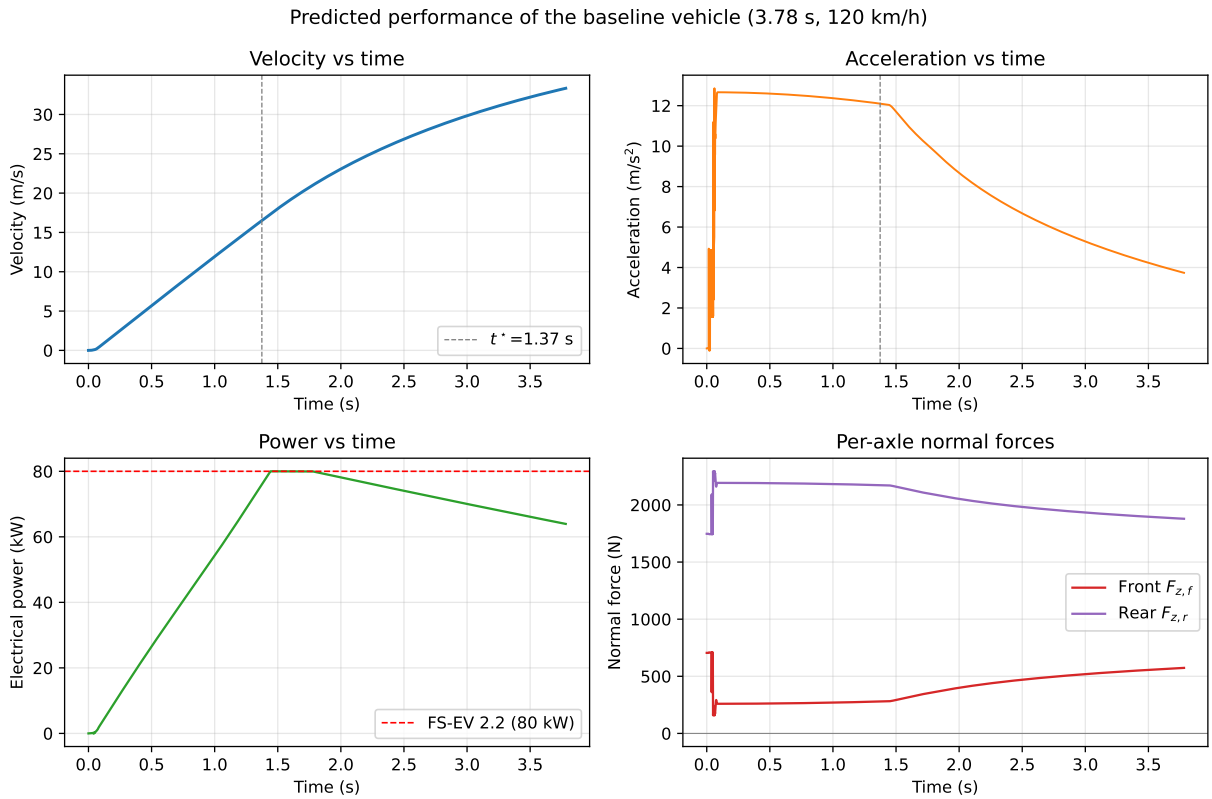


Figure 1.9: Predicted performance of the baseline vehicle. The grey dashed line at $t^\star = 1.37\,\mathrm{s}$ marks the transition from the traction-limited to the power-limited regime. Instantaneous electrical power (lower left) saturates cleanly at the FS-EV 2.2 cap, and the front normal force (lower right) remains positive throughout the run.

First, the vehicle spends $1.37\,\mathrm{s}$ ($36\,\%$ of the total run time) in the traction-limited regime and $2.41\,\mathrm{s}$ ($64\,\%$) in the power-limited regime, with the transition occurring at $v^\star = 15.4\,\mathrm{m\,s^{-1}}$ as predicted analytically in Section 1.2.4. The grip utilisation during the traction-limited phase averages approximately $103\,\%$ relative to the reference Pacejka peak, confirming the launch controller (Section 1.2.5) tracks the load-dependent grip ceiling closely.

Second, the electrical power saturates cleanly at $79.80\,\mathrm{kW}$ (against the $80\,\mathrm{kW}$ cap), without overshoot, throughout the power-limited phase. This is the evidence that Eq. 1.2.11 is correctly enforced at the electrical side rather than at the wheel.

Third, the front normal force falls smoothly from a static $707\,\mathrm{N}$ to a minimum of $158\,\mathrm{N}$ at peak acceleration and recovers as the vehicle enters the power-limited phase. The $158\,\mathrm{N}$ minimum is a $22\,\%$ fraction of the static load, which is ample margin against the wheelie threshold and validates the choice of anti-squat and CG geometry for the baseline configuration.

1.5.4 Compliance and predicted score

Rule compliance is evaluated automatically by the Rules layer:

- **FS-EV 2.2** — peak electrical power $P_{\text{peak}} = 79.80\,\text{kW} \leq 80\,\text{kW}$: **compliant.**
- **FS-D 5.3.1** — predicted time $t_{75} = 3.78\,\text{s} < 25\,\text{s}$: **compliant.**
- **Wheelie check** — minimum front normal force $\min F_{z,f} = 158\,\text{N} > 0$: **compliant.**

The acceleration-event score is computed from FS-D 5.3.2 as

$$M = 0.95\,P_{\text{max}}\,\frac{T_{\text{max}}/T_{\text{team}} - 1}{0.5} + 0.05\,P_{\text{max}}, \qquad T_{\text{max}} = 1.5\,T_{\text{fastest}}, \tag{1.5.1}$$

with $P_{\text{max}} = 75$ points. Taking $T_{\text{fastest}} = 3.5\,\text{s}$ as a representative competition-winning time and $T_{\text{team}} = 3.78\,\text{s}$ gives a predicted score of 59.2 points, rising to 65.1 points at the optimised $3.67\,\text{s}$.

Table 1.11: Requirements-to-evidence matrix for the baseline configuration. Each Formula Student requirement is evaluated in the cited subsection of this chapter, with the numerical evidence and compliance outcome.

Requirement	Rule	Evidence (§)	Value	Status
Electrical power $\leq 80\,\text{kW}$	FS-EV 2.2	1.5.4	$79.80\,\text{kW}$	Pass
Finish time $< 25\,\text{s}$	FS-D 5.3.1	1.5.4	$3.78\,\text{s}$	Pass
No front-axle lift	–	1.5.3	$158\,\text{N}$	Pass
Predicted acceleration score (FS-D 5.3.2)	FS-D 5.3.2	1.5.4	59.2 / 75 points	–

Section Contribution: Interim conclusion. The baseline vehicle is predicted to complete the event in $3.78\,\text{s}$, reaching $33\,\text{m}\,\text{s}^{-1}$, at a peak electrical power of $79.80\,\text{kW}$ with a minimum front normal force of $158\,\text{N}$. All three compliance checks pass, and the predicted FS-D 5.3.2 score is 59.2 of 75 points. The optimised configuration raises this to $3.67\,\text{s}$ and 65.1 points while remaining compliant.

1.6 Discussion and Limitations

The model captures longitudinal dynamics, load transfer with anti-squat, load-sensitive Pacejka tyre grip with an optional thermal overlay, closed-loop traction-controlled launch, electrical-side power limiting, aerodynamic drag and downforce, and rule compliance with scoring. Each is validated in Section 1.3. Four effects are currently neglected; each is discussed below with an estimate of its direction and magnitude on the predicted $75\,\text{m}$ time.

Driveline torsional compliance. A real driveline has a finite torsional stiffness that introduces a low-frequency oscillation in wheel torque following a step change (for example at the release of the torque ramp). For the single-ratio 5.5 reduction and a representative half-shaft stiffness of $\sim$

$5\,\mathrm{kN\,m\,rad^{-1}}$, the natural frequency is $25\,\mathrm{Hz}$, well above the launch-controller bandwidth but with the potential to add $20\,\mathrm{ms}$ of launch-transient losses. Including a two-mass torsional driveline model in the state vector would make this quantitative; the expected direction is a small increase in t_{75}.

Transient motor electromagnetic dynamics. The model assumes instantaneous torque response to a current request. The YASA P400R has an electrical time constant of $\sim 2\,\mathrm{ms}$, commensurate with the simulation timestep; field-weakening transitions above $70\,\%$ peak speed introduce additional torque dip of order $5\,\%$ for $\sim 50\,\mathrm{ms}$ [4]. The expected direction on t_{75} is a small increase.

Accumulator voltage sag. The model uses a fixed $V_{\text{bus}} = 600\,\mathrm{V}$, whereas the supercapacitor pack voltage falls as energy is drawn. With the pack capacitance of $3\,\mathrm{F}$ (Paul's chapter), a full-power draw for $3.8\,\mathrm{s}$ corresponds to a terminal-voltage drop from $600\,\mathrm{V}$ to approximately $510\,\mathrm{V}$. As the power limit is enforced on $V_{\text{bus}} I_m$, the simulator and the real system produce the same electrical power; the difference is a higher current demand in reality, which marginally increases ohmic losses inside the pack. Coupling with Paul's voltage-sag model is the most important extension and is the first item of further work.

Driver reaction time. The sim starts at $t = 0$ with the vehicle at rest and the throttle open. Published FS-E acceleration times include a $0.2\,\mathrm{s}$ to $0.4\,\mathrm{s}$ reaction penalty from the start signal; adding this to the predicted $3.78\,\mathrm{s}$ brings the total expected run time to $4.0\,\mathrm{s}$ to $4.2\,\mathrm{s}$, consistent with competitive but not record-setting times.

Combining the Monte Carlo parametric envelope (Section 1.4.2, to be finalised) with the estimated unmodelled effects above gives an overall confidence interval of approximately $\pm 0.20\,\mathrm{s}$ on the headline prediction.

1.7 Conclusions and Further Work

The following conclusions are drawn from the evidence presented in Sections 1.3–1.6:

- A modular longitudinal-dynamics simulation has been developed for the Formula Student acceleration event. The model includes quasi-static load transfer, load-sensitive Pacejka tyre grip with an optional thermal overlay, closed-loop traction-controlled launch, and electrical-side $80\,\mathrm{kW}$ power limiting, and is integrated by fourth-order Runge–Kutta at $1\,\mathrm{ms}$.

- The model is validated within $1\,\%$ of steady-state hand calculations, within $0.08\,\%$ of the limiting continuous-time integration result, and $40\,\%$ above the analytical constant-power lower bound. (Section 1.3.)

- Performance is dominated by longitudinal CG position and tyre peak grip; drivetrain efficiency, gear ratio, mass and wheel radius form a second-order band; aerodynamic drag and rolling resistance are negligible over $75\,\mathrm{m}$. (Section 1.4.1.)
- Numerical optimisation over seven independent decision variables, subject to the FS-EV 2.2, FS-D 5.3.1, and pitch-stability constraints, converges in 271 Nelder–Mead evaluations to a design predicted at $3.67\,\mathrm{s}$ for the $75\,\mathrm{m}$ event — a $2.9\,\%$ improvement on the baseline. (Section 1.4.3.)
- The design remains compliant under a $40\,\%$ surface-grip reduction, with a $28\,\%$ time penalty and no violation of power or time limits. (Section 1.4.4.)
- The baseline configuration is predicted to score 59.2 of 75 points under FS-D 5.3.2, rising to 65.1 at the optimised parameter set (Section 1.5.4).

Further work

- **Pacejka calibration with measured tyre data.** The current Pacejka coefficients are derived from lateral Avon FSAE data and representative FSAE longitudinal literature. Procurement of rig-test data would reduce the single largest source of parametric uncertainty identified in Section 1.4.1.
- **Coupled accumulator voltage sag.** The simulation currently assumes a fixed V_{bus}. Coupling with the voltage-sag model of the Supercapacitors chapter (Paul Vogelberg) would capture the interaction between pack ESR, terminal voltage and the active power limit near peak current.
- **Driveline torsional compliance.** Adding a two-mass torsional driveline to the state vector would quantify the launch-transient losses currently absorbed into the $80\,\mathrm{ms}$ torque ramp.
- **Transient motor thermal and electromagnetic dynamics.** Including motor thermal derating and field-weakening transitions would narrow the residual gap between the simulated acceleration time and the measured performance of real Formula Student Electric vehicles.
- **Extension to further events.** The same modular framework could be re-scoped to Skidpad and Autocross to verify that the acceleration-event optimum does not penalise overall Formula Student championship score.

References

[1] Hans B. Pacejka. *Tyre and Vehicle Dynamics*. 3rd. Oxford: Butterworth-Heinemann, 2012. ISBN: 978-0080970165.

[2] William F. Milliken and Douglas L. Milliken. *Race Car Vehicle Dynamics*. Warrendale, PA: Society of Automotive Engineers, 1995. ISBN: 978-1560915263.

[3] *BAMOCAR PG-D3 700/400 Motor Controller – Technical Datasheet*. Supplied with the project repository. Unitek Industrie-Elektronik GmbH. [PH year accessed]. URL: %5BPH:%20manufacturer%20URL%5D.

[4] *YASA P400 R Series Axial Flux Motor – Product Datasheet*. Revision 14, supplied with the project repository. YASA Ltd. [PH year accessed]. URL: %5BPH:%20manufacturer%20URL%5D.

[5] William H. Press et al. *Numerical Recipes: The Art of Scientific Computing*. 3rd. Cambridge: Cambridge University Press, 2007. ISBN: 978-0521880688.